# fish_biomass_NCRMP

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## 2. Download Data

```
# Get all USVI data for specified years and regions
USVI <- getRvcData(years = c(2017:2025), regions = c("STTSTJ", "STX"))
```

### Cunstruct species lookup table

You can also embed plots, for example:

```
##   SPECIES_CD        FAMILY                    SCINAME          COMNAME LC LM
## 1   ABL HIAN     Belonidae             Ablennes hians  flat needlefish NA NA
## 2   ABU SAXA Pomacentridae       Abudefduf saxatilis   sergeant major NA NA
## 3   ABU TAUR Pomacentridae          Abudefduf taurus   night sergeant NA NA
## 4   ACA ASPE  Chaenopsidae   Acanthemblemaria aspera roughhead blenny NA NA
## 5   ACA CHAP  Chaenopsidae Acanthemblemaria chaplini papillose blenny NA NA
## 6   ACA MARI  Chaenopsidae    Acanthemblemaria maria secretary blenny NA NA
##     WLEN_A WLEN_B
## 1       NA     NA
## 2 1.64e-05  3.142
## 3       NA     NA
## 4 8.40e-06  2.962
## 5 8.40e-06  2.962
## 6 8.40e-06  2.962
```

```
# 5. Estimate Biomass at PSU Level for All Species ----------------------------
# Calculate mean biomass per PSU per species (no group parameter, for species-level)
gpsu_bio_species <- getPSUBiomass(
  USVI,
  species = group_df2$SPECIES_CD
)

# 6. Add Group & Environmental Data -------------------------------------------
# Attach group and species names, for reference and aggregation
gpsu_bio_species <- gpsu_bio_species %>%
  left_join(group_df2 %>% select(SPECIES_CD, group, SPECIES), by = "SPECIES_CD")

# Create a spatial/environmental lookup table (unique per PSU)
psu_env <- USVI$sample_data %>%
```

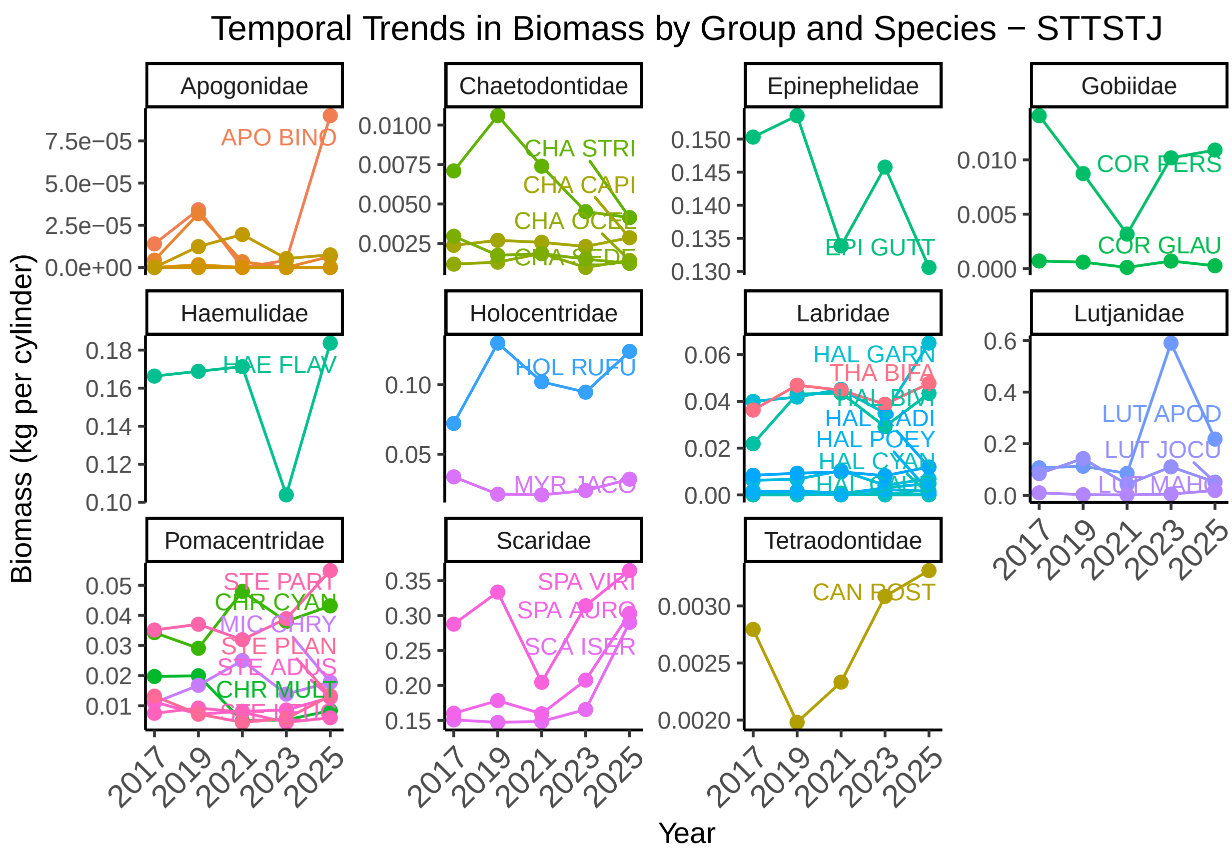
```r
  select(PRIMARY_SAMPLE_UNIT, LAT_DEGREES, LON_DEGREES, DEPTH) %>%
  distinct(PRIMARY_SAMPLE_UNIT, .keep_all = TRUE) # ensures one row per PSU

# Join spatial/environmental info
final_gpsu <- gpsu_bio_species %>%
  left_join(psu_env, by = "PRIMARY_SAMPLE_UNIT")

# 7. Keep and Order Only Needed Columns ---------------------------------------
final_gpsu <- final_gpsu %>%
  select(
    YEAR, REGION, STRAT, PROT, PRIMARY_SAMPLE_UNIT,
    SPECIES, SPECIES_CD, group, m, var, biomass,
    LAT_DEGREES, LON_DEGREES, DEPTH
  )

head(final_gpsu)
```

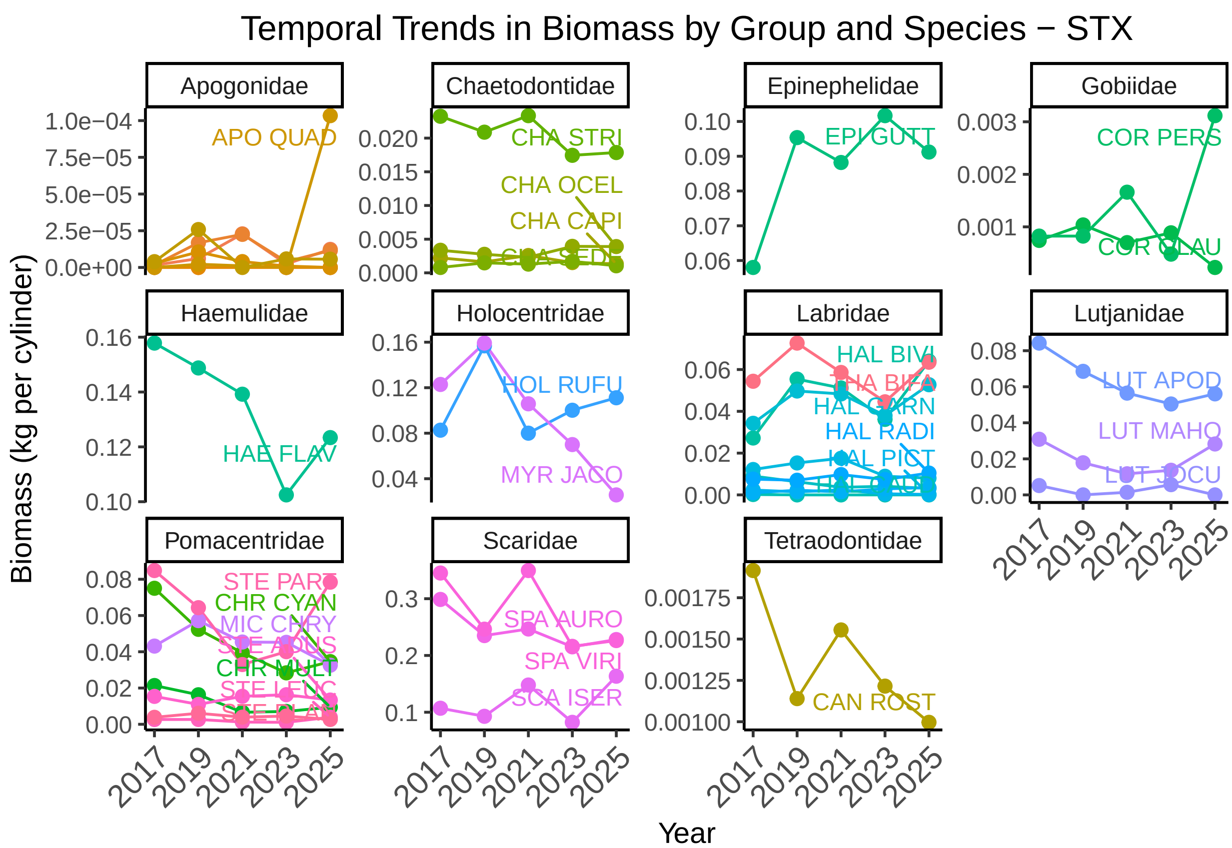
```
##   YEAR REGION    STRAT PROT PRIMARY_SAMPLE_UNIT                 SPECIES
## 1 2017 STTSTJ AGRFDEEP    0                6018    Apogon aurolineatus
## 2 2017 STTSTJ AGRFDEEP    0                6018       Apogon binotatus
## 3 2017 STTSTJ AGRFDEEP    0                6018       Apogon maculatus
## 4 2017 STTSTJ AGRFDEEP    0                6018          Apogon phenax
## 5 2017 STTSTJ AGRFDEEP    0                6018 Apogon pseudomaculatus
## 6 2017 STTSTJ AGRFDEEP    0                6018 Apogon quadrisquamatus
##   SPECIES_CD      group m var biomass LAT_DEGREES LON_DEGREES DEPTH
## 1   APO AURO Apogonidae 1  NA       0    18.26901    -64.7264    27
## 2   APO BINO Apogonidae 1  NA       0    18.26901    -64.7264    27
## 3   APO MACU Apogonidae 1  NA       0    18.26901    -64.7264    27
## 4   APO PHEN Apogonidae 1  NA       0    18.26901    -64.7264    27
## 5   APO PSEU Apogonidae 1  NA       0    18.26901    -64.7264    27
## 6   APO QUAD Apogonidae 1  NA       0    18.26901    -64.7264    27
```

# Plot

```r
#Summarize for Visualization (Species-Level within Groups) ----------
plot_bio <- final_gpsu %>%
  group_by(REGION, YEAR, group, SPECIES_CD) %>%
  summarise(
    mean_biomass = mean(biomass, na.rm = TRUE),
    .groups = "drop"
  )

# 2. Loop by region and plot one at a time
for (reg in unique(plot_bio$REGION)) {
  plot_bio_reg <- plot_bio %>% filter(REGION == reg)

  # Prepare label data (last year for each group/species)
  label_data_reg <- plot_bio_reg %>%
    group_by(group, SPECIES_CD) %>%
    filter(YEAR == max(YEAR)) %>%
```

```r
    ungroup()

  p <- ggplot(plot_bio_reg, aes(x = YEAR, y = mean_biomass, color = SPECIES_CD, group = SPECIES_CD)) +
  geom_line() +
  geom_point(size = 2) +
  ggrepel::geom_text_repel(
    data = label_data_reg,
    aes(label = SPECIES_CD, color = SPECIES_CD),
    nudge_x = 0.2,
    direction = "y",
    hjust = 0,
    show.legend = FALSE,
    size = 3
  ) +
  scale_x_continuous(breaks = c(2017, 2019, 2021, 2023, 2025)) +
  facet_wrap(~ group, scales = "free_y") +
  theme_classic() +
  ylab("Biomass (kg per cylinder)") +
  xlab("Year") +
  ggtitle(paste("Temporal Trends in Biomass by Group and Species -", reg)) +
  theme(
    plot.title    = element_text(hjust = 0.5),
    legend.position = "none",
    axis.text.x   = element_text(angle = 45, hjust = 1, size = 12) # add this line!
  )

  print(p)
  # Optionally, save:
  # ggsave(paste0("Temporal_Trends_", reg, ".png"), plot = p, width = 10, height = 7)
}
```


Temporal Trends in Biomass by Group and Species − STTSTJ
Apogonidae
Chaetodontidae
Epinephelidae
Gobiidae
Haemulidae
Holocentridae
Labridae
Lutjanidae
Pomacentridae
Scaridae
Tetraodontidae
APO BINO
CHA STRI
CHA CAPI
CHA OCEL
CHA SEDE
EPI GUTT
COR PERS
COR GLAU
HAE FLAV
HOL RUFU
MYR JACO
HAL GARN
THA BIFA
HAL BIVI
HAL RADI
HAL POEY
HAL CYAN
LUT APOD
LUT JOCU
LUT MAHO
STE PART
CHR CYAN
MIC CHRY
STE PLAN
STE ADUS
CHR MULT
SPA VIRI
SPA AURO
SCA ISER
CAN ROST
Biomass (kg per cylinder)
Year

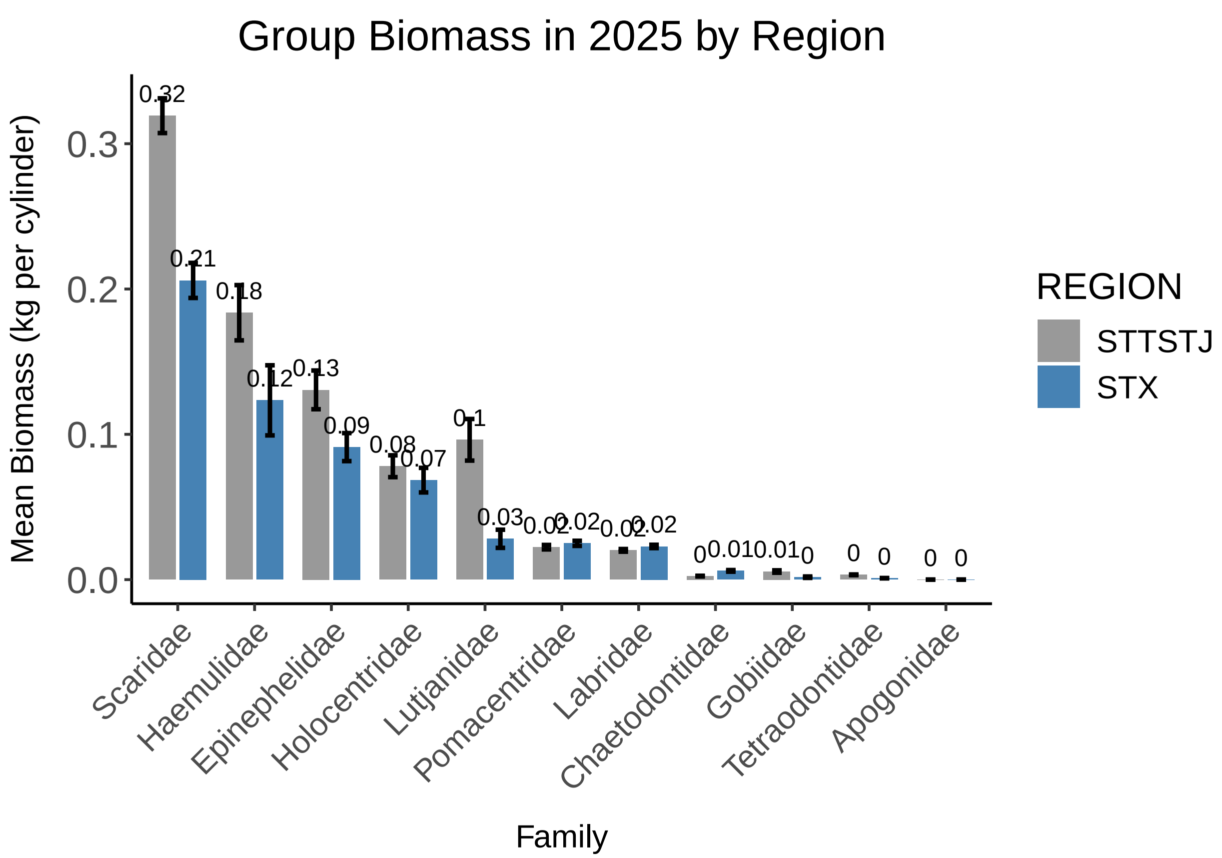



### General plot by group

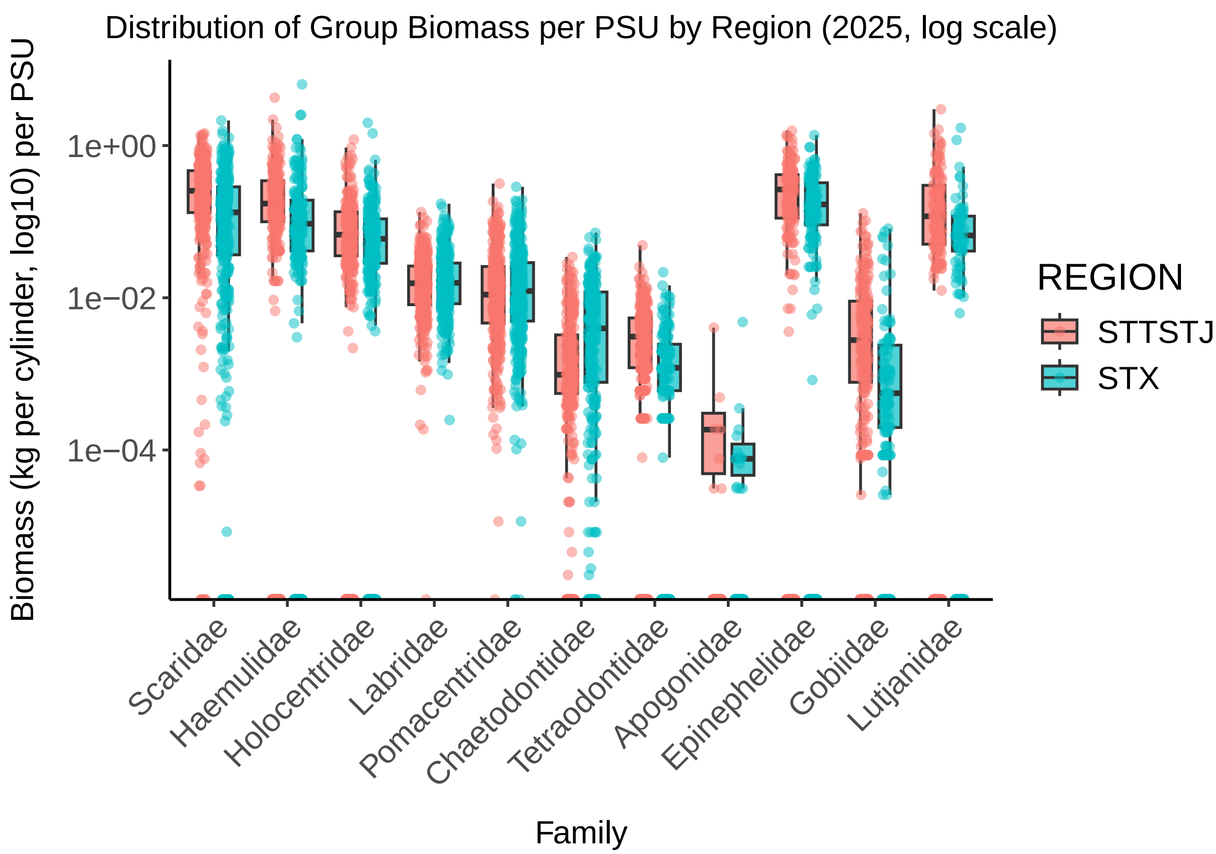
### YEAR 2025

```r
# 1. Group by REGION and group for 2025
bar_group <- final_gpsu %>%
  filter(YEAR == 2025) %>%
  group_by(REGION, group) %>%
  summarise(
    mean_biomass = mean(biomass, na.rm = TRUE),
    se_biomass   = sd(biomass, na.rm = TRUE) / sqrt(sum(!is.na(biomass))),
    .groups      = "drop"
  )

# For sorting and clear side-by-side plot, arrange group factor by overall mean across regions
bar_group <- bar_group %>%
  group_by(group) %>%
  mutate(overall_mean = mean(mean_biomass, na.rm=TRUE)) %>%
  ungroup() %>%
  mutate(group = reorder(group, -overall_mean))

p_bar_biomass <- ggplot(bar_group, aes(x = group, y = mean_biomass, fill = REGION)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_errorbar(
```

```
    aes(ymin = mean_biomass - se_biomass, ymax = mean_biomass + se_biomass, group = REGION),
    position = position_dodge(width = 0.8),
    width = 0.25, size = 0.8
  ) +
  geom_text(
    aes(label = round(mean_biomass, 2), group = REGION),
    position = position_dodge(width = 0.8),
    vjust = -0.8, size = 3
  ) +
  theme_classic() +
  ylab("Mean Biomass (kg per cylinder)") +
  xlab("Family") +
  ggtitle("Group Biomass in 2025 by Region") +
  scale_fill_manual(values = c("grey60", "steelblue")) + # Or pick 2 colors you like
  theme(
    plot.title   = element_text(hjust = 0.5, size = 16),
    axis.text.x  = element_text(angle = 45, hjust = 1, size = 12),
    axis.text.y  = element_text(size = 14),
    axis.title.x = element_text(size = 12, margin = margin(t = 10)),
    axis.title.y = element_text(size = 12, margin = margin(r = 10)),
    legend.title = element_text(size = 14),
    legend.text  = element_text(size = 12)
  )
p_bar_biomass
```



```r
# Aggregate: one value per REGION, group, and PSU (mean biomass of group per PSU for that region)
box_data <- final_gpsu %>%
  filter(YEAR == 2025) %>%
  group_by(REGION, group, PRIMARY_SAMPLE_UNIT) %>%
  summarise(
    group_biomass = mean(biomass, na.rm = TRUE),
    .groups = "drop"
  )

# Order groups by median biomass across all regions, for left-to-right order
# (Same order for both regions, helps comparison)
median_group <- box_data %>%
  group_by(group) %>%
  summarise(med = median(group_biomass, na.rm = TRUE)) %>%
  arrange(desc(med))
box_data$group <- factor(box_data$group, levels = median_group$group)

p_box_biomass <- ggplot(box_data, aes(x = group, y = group_biomass, fill = REGION)) +
  geom_boxplot(
    outlier.shape = NA,
    position = position_dodge(width = 0.8),
    width = 0.6,
    alpha = 0.7
  ) +
  geom_jitter(
    aes(color = REGION),
    size = 1.2,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.12, dodge.width = 0.6)
  ) +
  scale_y_log10() +
  ylab("Biomass (kg per cylinder, log10) per PSU") +
  xlab("Family") +
  ggtitle("Distribution of Group Biomass per PSU by Region (2025, log scale)") +
  theme_classic() +
  theme(
    plot.title   = element_text(hjust = 0.5, size = 12),
    axis.text.x  = element_text(angle = 45, hjust = 1, size = 12),
    axis.text.y  = element_text(size = 12),
    axis.title.x = element_text(size = 12, margin = margin(t = 10)),
    axis.title.y = element_text(size = 12, margin = margin(r = 10)),
    legend.title = element_text(size = 14),
    legend.text  = element_text(size = 12)
  )
p_box_biomass
```




```r
# New name for the boxplot data to avoid confusion
box_data_species <- final_gpsu %>%
  filter(YEAR == 2025) %>%
  group_by(group) %>%
  mutate(SPECIES_CD = reorder(SPECIES_CD, -biomass, FUN = median, na.rm = TRUE)) %>%
  ungroup()

p_box_biomass_sp <- ggplot(box_data_species, aes(x = SPECIES_CD, y = biomass, fill = REGION)) +
  geom_boxplot(
    outlier.shape = NA,
    position = position_dodge(width = 0.75),
    width = 0.6,
    alpha = 0.7
  ) +
  geom_jitter(
    aes(color = REGION),
    size = 1.2,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.10, dodge.width = 0.6)
  ) +
  scale_y_log10() +
  ylab("Biomass (kg per cylinder, log10)") +
  xlab("Species") +
  ggtitle("Distribution of Biomass by Species and Region Within Groups (2025)") +
  facet_wrap(~ group, scales = "free_x") +
  theme_classic() +
```

```
  theme(
    plot.title     = element_text(hjust = 0.5, size = 18),
    strip.text     = element_text(size = 14),
    axis.text.x    = element_text(angle = 45, hjust = 1, size = 11),
    axis.title.x   = element_text(size = 16, margin = margin(t = 10)),
    axis.title.y   = element_text(size = 16, margin = margin(r = 10)),
    axis.text.y    = element_text(size = 14),
    legend.title   = element_text(size = 13),
    legend.text    = element_text(size = 12)
  )
print(p_box_biomass_sp)
```